

Characterization of high-Cr white cast iron reinforced with Ti-Al-C, Ti-Ni-C & Ti-Ni-B₄C

Diogo Mota

Departamento de Engenharia Metalúrgica e de Materiais, Faculdade de Engenharia, Universidade do Porto, Rua Dr. Roberto Frias, 4200-465 PORTO, Portugal (up201704847@up.pt) ORCID [0009-0001-1554-7389](https://orcid.org/0009-0001-1554-7389)

Ilídio Costa

Departamento de Engenharia Metalúrgica e de Materiais, Faculdade de Engenharia, Universidade do Porto, Rua Dr. Roberto Frias, 4200-465 PORTO, Portugal (up202006411@up.pt) ORCID [0009-0003-0793-4180](https://orcid.org/0009-0003-0793-4180)

Bruno Cunha

Departamento de Engenharia Metalúrgica e de Materiais, Faculdade de Engenharia, Universidade do Porto, Rua Dr. Roberto Frias, 4200-465 PORTO, Portugal (up202004620@up.pt) ORCID [0009-0004-0580-3758](https://orcid.org/0009-0004-0580-3758)

Aida B. Moreira

Departamento de Engenharia Metalúrgica e de Materiais, Faculdade de Engenharia, Universidade do Porto and LAETA/INEGI – Institute of Science and Innovation in Mechanical and Industrial Engineering, Rua Dr. Roberto Frias, 4200-465 PORTO, Portugal (aidamoreira@fe.up.pt) ORCID [0000-0001-8615-7612](https://orcid.org/0000-0001-8615-7612)

Author Keywords

High-chromium white cast iron, local reinforcement, metal matrix composite, microstructural characterization.

Type: Research Article

 Open Access

 Peer Reviewed

 CC BY

Abstract

High-chromium white cast iron (HCrWCI) specimens locally reinforced with WC-metal matrix composites were produced via an in-situ technique: powder mixtures of the systems Ti-Al-C, Ti-Ni-C and Ti-Ni-B₄C were cold-pressed in a pre-form were inserted in the mold cavity before pouring the base metal. The microstructure of the Ti-Al-C system resulting reinforcement is a matrix of martensite (α') with TiC particles and M₇C₃ carbides, with a hardness of 892±59 HV. In the Ti-Ni-C system reinforcement a matrix of austenite (γ) with TiC particles and M₇C₃ carbides was seen, with a hardness of 973±55 HV. In the Ti-Ni-B₄C system reinforcement, an γ matrix with TiC and TiB₂ particles, and (Cr,Fe)₂B resulting in a hardness of 965±65 HV. The Ni present in the Ti-Ni-C and Ti-Ni-B₄C contributed to the γ stabilization explaining these systems possessing an γ matrix instead of the matrix of α' seen in the Ti-Al-C system. The Ti-Al-C system had a brittle layer of alumina (Al₂O₃) in the interface between the reinforcement and the base metal with a lot of porosity, while the systems with Ni a reaction zone was present in the interface, improving the bonding.

1. Introduction

High chromium white cast irons (HCrWCIs) have shown efficacy as a material suitable for a plethora of applications within aggressive environments necessitating diverse resistances to abrasion, erosion, and erosion-corrosion (Chung et al. 2013; Karantzalis, Lekatou, et Mavros 2013). In an effort to increase the wear resistance, several reinforcements have been studied (Moreira et al. 2022; Moreira, Ribeiro, et al. 2020; Moreira, Sousa, et al. 2020). HCrWCIs metal matrix composites reinforced with Ti-Al-C, Ti-Ni-C & Ti-Ni-B₄C in situ systems will be studied and characterized.

HCrWCI's exceptional wear resistance is attributed to a combination of hard eutectic and primary carbides (M_7C_3) and a ferrous matrix, which could be austenite (γ), martensite (α'), ferrite (δ). As cast HCrWCIs usually exhibit an γ matrix, although ferrite has also been found, associated with high cooling rates. After a destabilization heat treatment, aimed at increasing the wear resistance (Gonzalez-Pociño, Alvarez-Antolin, et Asensio-Lozano 2019), a secondary precipitation of fine $M_{23}C_6$ carbides occurs coupled with the transformation of γ matrix into α' (Doğan, Hawk, et Laird 1997; Karantzalis, Lekatou, et Mavros 2013; Chung et al. 2013; Wiengmoon, Chairuang Sri, et Pearce 2005). The chemical composition of the HCrWCI studied can be seen in **Table 1**, being classified as a class III and type A abrasion-resistant cast iron following ASTM A532 (ASTM 2019).

Table 1: Chemical composition of HCrWCI alloy.

Cr	C	Ni	Mn	Si	Fe
26.80	3.10	0.30	0.54	0.60	Balance

In situ reinforcement methods aim to achieve hard ceramic particles through an exothermal reaction which is activated by the heat of the liquid metal when in contact with the compacts. Self-propagating high-temperature synthesis (SHS) is a popular technology to achieve in situ ceramic particles due to its high time efficiency and product purity. However, it should be noted that final products can possess porosities. This methodology allows for the local reinforcement of the casting, instead of a homogeneous distribution throughout the body (Y.-f. Yang et al. 2007). TiC and TiB_2 are the within the best ferrous matrix reinforcements, owing to their good wettability, high hardness, stability in ferrous matrixes, and excellent wear resistance, so the SHS systems studied were aim at obtaining these particles (Zou et al. 2012; Wang et al. 2013).

Analyzing the Ti-Al-C system, several phases can be found. During the combustion reaction, Al reacts with the Ti to form $TiAl_x$, which later react the C and decomposed into TiC and Al (Choi et Rhee 1993). The introduction of Al to the Ti-C decreases the combustion temperature and size of TiC particles decrease (Song et al. 2009). Moreira et al. found that titanium formed mostly TiC and that the aluminum atoms were in a solid solution of α' , as well as, reacting with oxygen present the molds and forming aluminum oxides (Moreira, Sousa, et al. 2020).

Akin to the Ti-Al-C, in the Ti-Ni-C system the Ni decreases the combustion temperature and the size of TiC particle, whilst altering their morphology from angular to spherical (Zhu et al. 2017; Jie-Cai, Zhang, et Wood 2000). Jiang et al. found TiC precipitates in the Fe matrix, and that Ni atoms dissolved into the Fe solid solution. Yang et al., in addition to these phases, also found TiNi (Y.-F. Yang, Wang, Liang, et al. 2008).

Along with TiC, the system Ti-Ni-B₄C aims to obtain TiB_2 particles. The Ni, similarly to the other system, decreases the combustion temperature, particle size and ignition time while forming other constituents at higher percentages due to incomplete reactions (Y.F. Yang, Wang, Zhao, et al. 2008). In addition to TiC and TiB_2 , M_2B might also be found when taking into consideration of the high percentage of Cr (Raghavan 2003; Bedolla-Jacuinde et al. 2021; Purba et al. 2022). The nickel atoms dissolve into the Fe matrix, stabilizing γ and thereby decreasing the amount of α' formed.

2. Materials and Methods

The production of the samples begun with the selection of the powders for each of the reinforcement system, the particle size distribution of each powder is present in **Table 2**.

Following the selection, the powder is properly weighted in proportions to match the formation of the desired hard precipitates for each system.

Table 2: Particle size distribution of each powder and composition of each system.

	Ti	Ni	Al	C _{graphite}	B ₄ C
D ₅₀ (μm)	41.87	12.13	10.93	40.42	33.20
D ₉₀ (μm)	68.17	29.97	21.48	80.89	56.70
Ti-Al-C (wt.%)	64	-	20	16	-
Ti-Ni-C (wt.%)	64	20	-	16	-
Ti-Ni-B ₄ C (wt.%)	50.5	30	-	-	19.5

Once the optimal proportion of element has been determined, the metal powder must be mixed to achieve a homogeneous microstructure. This process is typically facilitated by a Turbula shaker-mixer. Following this, the powders are thoroughly mixed, and the metallic powder system is compacted using a cold presser. Once the compaction stage is complete, the sample is positioned and fixed within the sand-casting mold. Finally, the cast iron is released from the mold and cut with wire EDM to form the final samples, as illustrated in **Figure 1**.

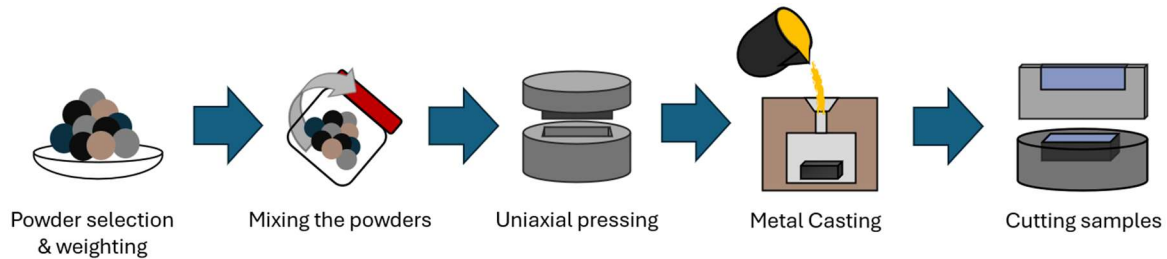


Figure 1: Illustration of the procedure to obtain the samples.

A microstructural characterization of reinforced and non-reinforced areas of the samples, as well as the interface region, to this effect, the metallographic preparation begun with grinding the samples, using two MD polishing cloths of 54 μm and 18 μm. Then, the samples were polished in polishing pads using diamond suspensions, 6 μm and 1 μm, respectively. Before the scanning electron microscope (SEM) analysis, a silica suspension of 0,06 μm was used. In addition to this the samples will be also analyzed with energy dispersive X-ray spectroscopy (EDS) and electron backscatter diffraction (EBSD).

Furthermore, Vickers macro and micro hardness tests were performed following ISO 6507-1:2023 (ISO 2023), and nano-hardness testing, ISO 14577-1:2015 (Standardization 2015). Wear resistance of the samples was evaluated through ball-catering following ISO 26424:2008 (Standardization 2008).

3. Results & Discussion

3.1. Base Metal

The unreinforced sections of all exhibit a microstructure that is identical to that of the base HCrWCI. As seen in

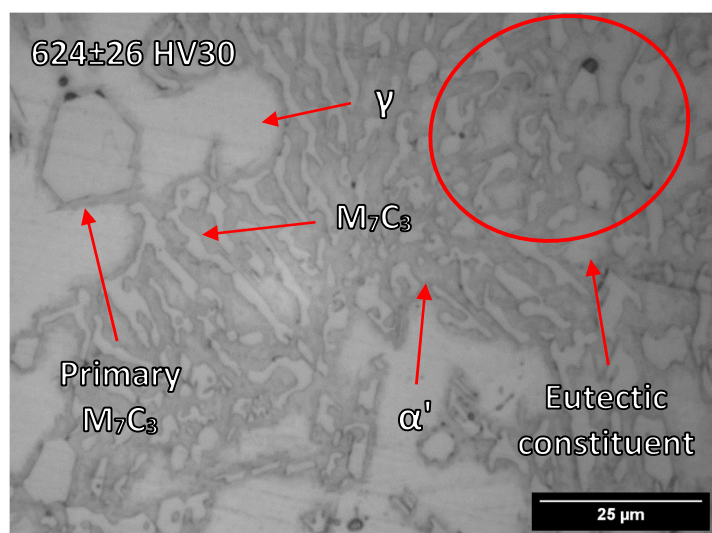


Figure 2: Microstructure of base HCrWCl, etched with Beraha-martensite reagent, in OM.

3.2. Characterization of Ti-Al-C reinforcement system

The profile of the reinforcement sample, perpendicular to the surface, reveals an uneven hardness throughout the reinforced region, possessing a higher hardness near the surface followed by a section highly heterogeneous region before reaching the base metal, as seen in **Figure 3**. The average macro hardness of the reinforced section is 892 ± 59 HV30 and 656 ± 11 HV30 on the base metal which is higher than the original alloy. These phenomena can be attributed to the heat of the exothermal reaction, which promoted diffusion of C and Cr into the carbides destabilized the γ matrix, and introduction of Al into the solid solution the Fe (Moreira, Sousa, et al. 2020).

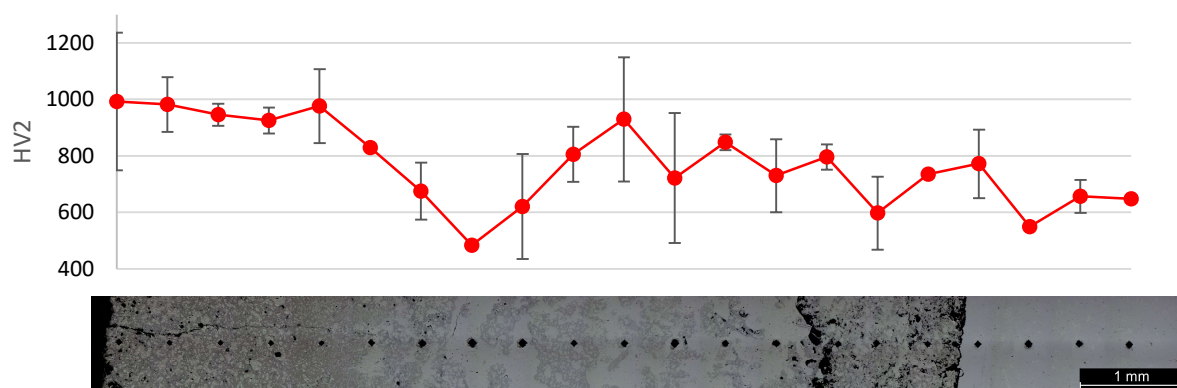


Figure 3: Hardness value throughout the profile of the Ti-Al-C sample.

As it is possible to note in **Figure 3** and more evidently in **Figure 4**, the interface between the base metal and reinforcement possesses some defects, such as pores, cracks and aluminum oxides, most like Al_2O_3 , brittle layer detailing the interface.

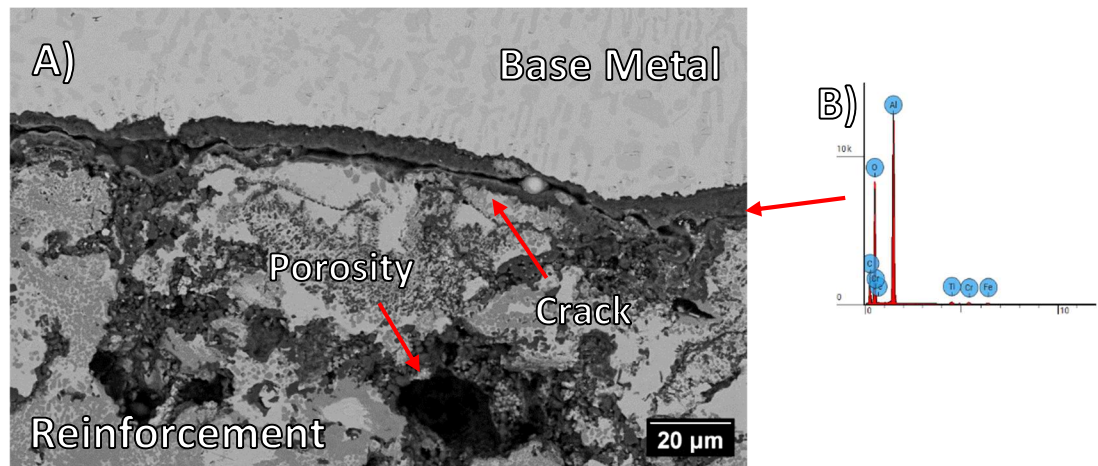


Figure 4: A) Microstructure of the interface between the reinforcement and base metal with backscattered electrons; B) EDS spectrum of the interface layer.

The bulk reinforcement possesses a heterogeneous microstructure, as seen in **Figure 5**, composed of regions of bare metal, and regions with TiC particles and a few aluminum oxides, most like Al_2O_3 (Moreira, Sousa, et al. 2020), in a matrix of α' and M_7C_3 , as its possible to conclude from the Kikuchi pattern in **Figure 5**. These oxides form due to interaction between Al and O for the air trapped inside the molding cavity.

The TiC precipitates, averaging $1.02 \pm 0.65 \mu\text{m}$, possess a high hardness, 3200 HV (Kopyciński et Piasny 2012), thereby increasing the overall hardness at the reinforced region. The ferrous matrix is α' , this can be attributed to the dissolution of Al into the matrix, which destabilizes the γ , coupled with the slower cooling rates due to the heat generated from the SHS reaction, promoting the diffusion of allowing elements such as C and Cr to form carbide (Moreira, Sousa, et al. 2020; Carpenter, Carpenter, et Pearce 2004). Olejnik et al. (Olejnik et al. 2011), owns the patchy nature of the reinforcement to the creation of “bubbles” due to the merging of single TiC crystal, filled with liquid metal.

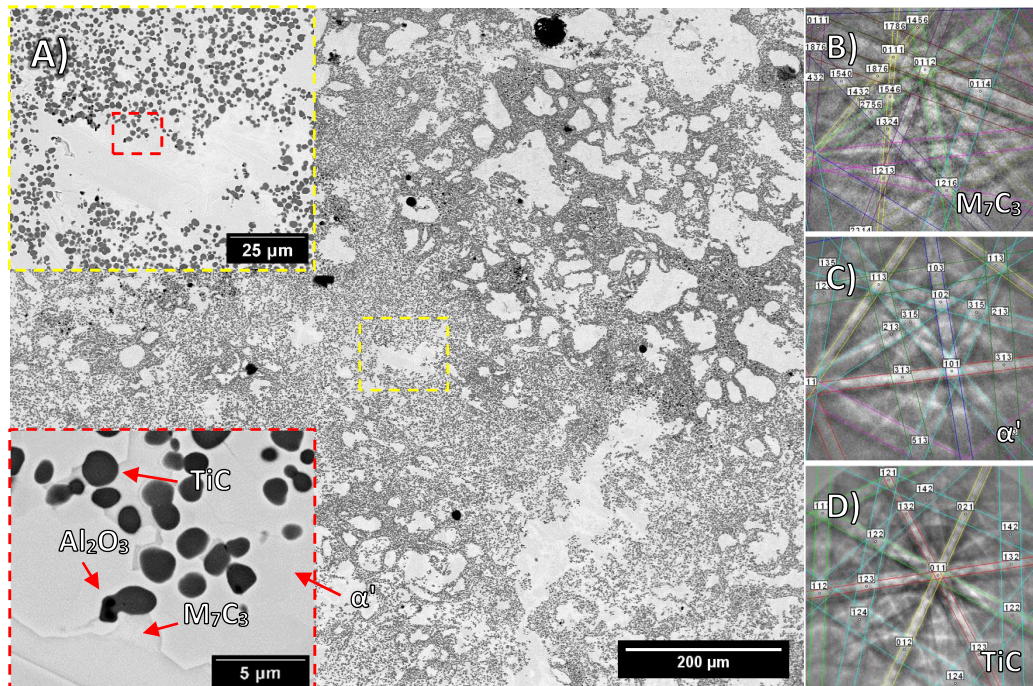


Figure 5: Sample of reinforced HCrWCI with Ti-Al-C; A) Microstructure of the reinforcement with back-scattered electrons; B) Kikuchi pattern of M_7C_3 ; C) Kikuchi pattern of α' ; D) Kikuchi pattern of TiC.

3.3. Characterization of Ti-Ni-C reinforcement system

The profile of the reinforcement sample, perpendicular to the surface, reveals an uneven hardness throughout the reinforced region, possessing a highly heterogeneous section owing to unreinforced layers that run parallel to the surface, before reaching the base metal, as seen in **Figure 6**.

The average macro hardness of the reinforced section is 973 ± 55 HV30 and 612 ± 26 HV30 on the base metal which is lower than the original alloy. These phenomena can be attributed to the addition of Ni which stabilizes γ during solidification, effectively reducing the α' which has a higher hardness than γ (Tabrett, Sare, et Ghomashchi 1996; Fashu et Trabadelo 2023).

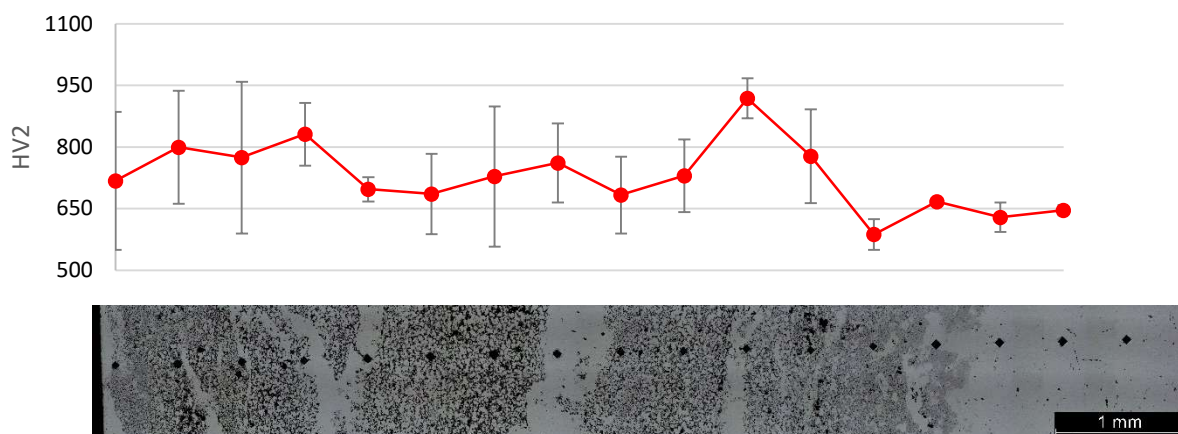


Figure 6: Hardness value throughout the profile of the Ti-Ni-C sample.

The unreinforced layers seen in **Figure 6** and more evidently in **Figure 7 A**, can be caused by the crack of the compact due to thermal shock or due to an inadequate compacting pressure, all system were compacted at the same pressure which may not be ideal due to the different compositions, but further investigation is needed. The interface between the base metal and reinforcement, as seen **Figure 7 B**, possess only a few porosities, with a reaction zone, showing good bonding without larger defects that could cause a discontinuity.

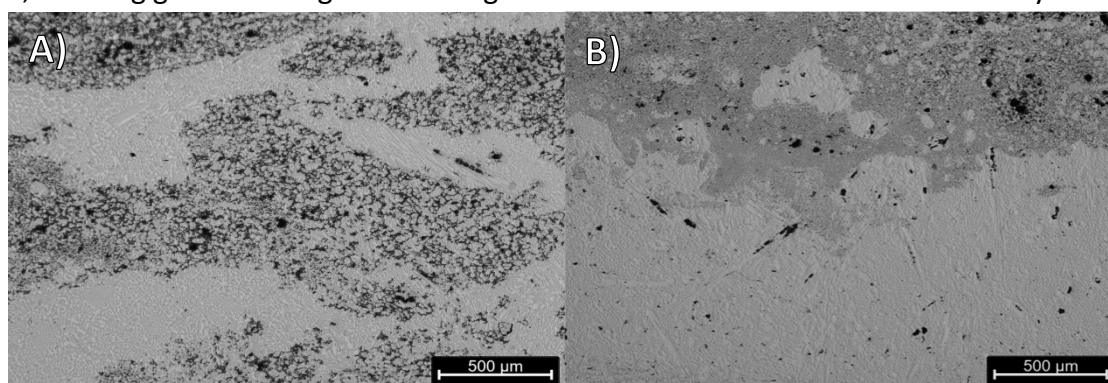


Figure 7: OM imaging of Ti-Ni-C sample; A) Layers in the reinforced region; B) interface between the base metal and reinforced region.

Analyzing the bulk of reinforcement, it possesses a heterogeneous microstructure composed of regions of base metal and regions with TiC particles, 3.09 ± 1.42 μm which tend to stick together as opposed to Ti-Al-C system which are dispersed in the matrix. This phenomenon might originate from increase of interface energy between liquid metal / TiC when compared to the Ti-Al-C system, so the single particles coalesce into the “sponge” like structure filled with liquid metal, as seen in **Figure 88 A**, to reduce the interface surface. These particles are suspended in M_7C_3 and γ matrix, as concluded by the Kikuchi pattern in **Figure 88 B, C and D**, owing to the addition of Ni which stabilizes γ by lowering the MS temperature.

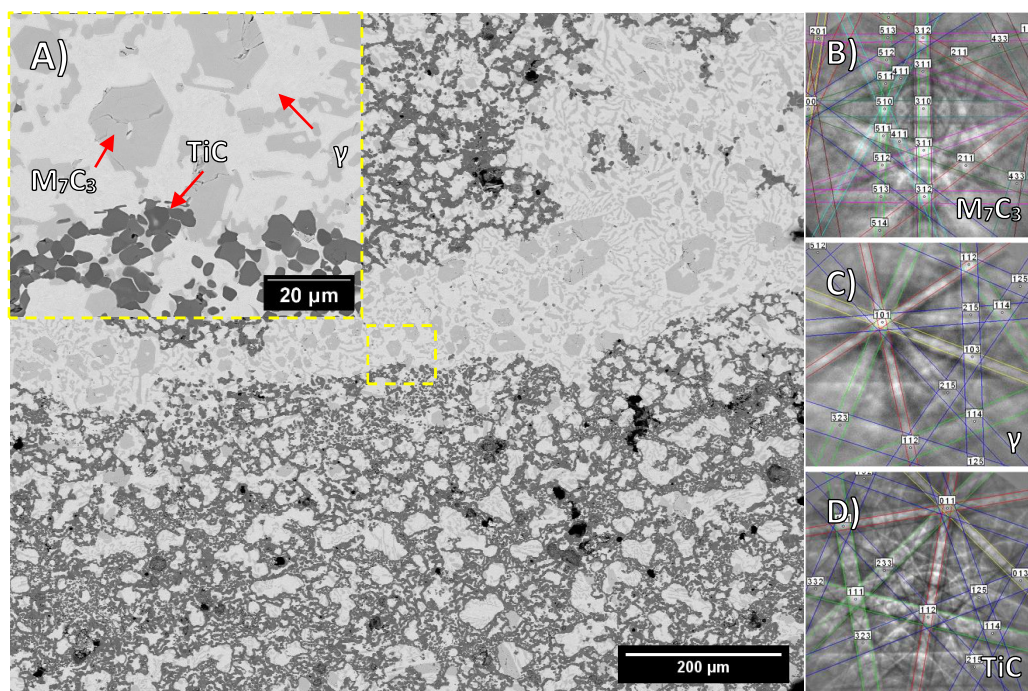


Figure 88: Sample of reinforced HCrWCI with Ti-Ni-C; A) Microstructure of the reinforcement with back-scattered electrons; B) Kikuchi pattern of M7C3; C) Kikuchi pattern of γ ; D) Kikuchi pattern of TiC.

Further analyzing the microstructure, it is possible to see the distribution of the different elements through EDS mapping in **Figure 99**. Cr, **Figure 99 B**, is mainly present in M_7C_3 whilst in smaller percentages in the γ matrix. Fe, **Figure 99 C**, mainly exists in the matrix, same with Ni, **Figure 99 D**, which is mostly dissolved in the matrix contributing to the γ stabilization. Si, **Figure 99 E**, is dispersed throughout all phases but at a higher concentration in the matrix. As expected, Ti, **Figure 99 F**, is almost only present in the TiC particles meaning all the titanium reacted with carbon to form these hard particles.

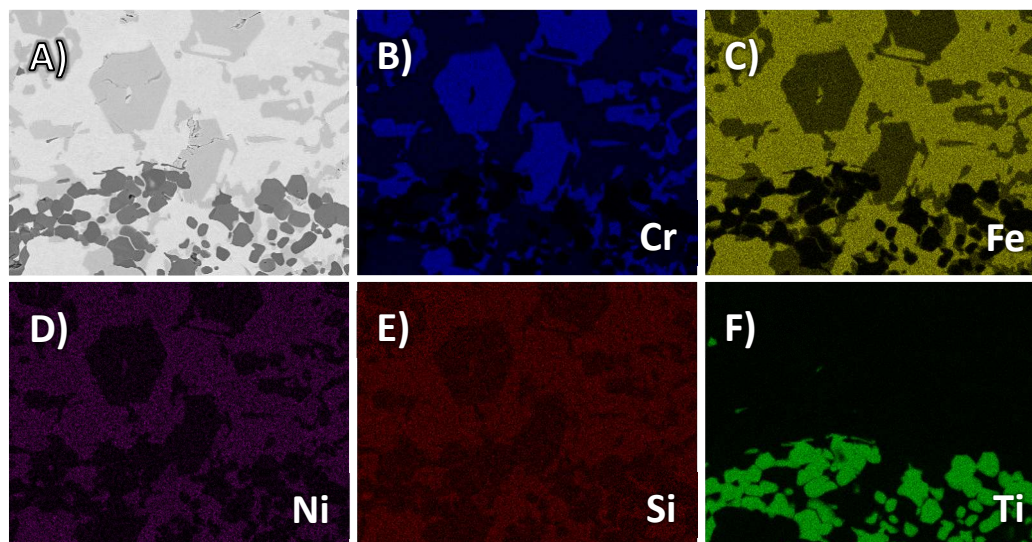


Figure 99: A) Back-scattered electron image; B) Cr EDS mapping; C) Fe EDS mapping; D) Ni EDS mapping; E) Si EDS mapping; F) Ti EDS mapping.

3.4. Characterization of Ti-Ni-B₄C reinforcement system

The profile of the reinforcement sample, perpendicular to the surface, reveals an uneven hardness throughout the reinforced region, possessing a lower hardness near the surface, followed by a highly heterogeneous and hard section, before reaching the base metal, as seen in **Figure 10 10**.

The average macro hardness of the reinforced section is 965 ± 65 HV30 and 567 ± 17 HV30 on the base metal which is lower than the original alloy. These phenomena can be attributed to the formation of $(Cr,Fe)_2B$, which since the compact were weight for a ratio of 1:2 of TiC:TiB₂, leads to the depletion of Cr an increase of C which stabilizes γ . The addition of Ni further stabilizes γ during solidification, effectively reducing the α' which has a higher hardness (Tabrett, Sare, et Ghomashchi 1996; Fashu et Trabadelo 2023).

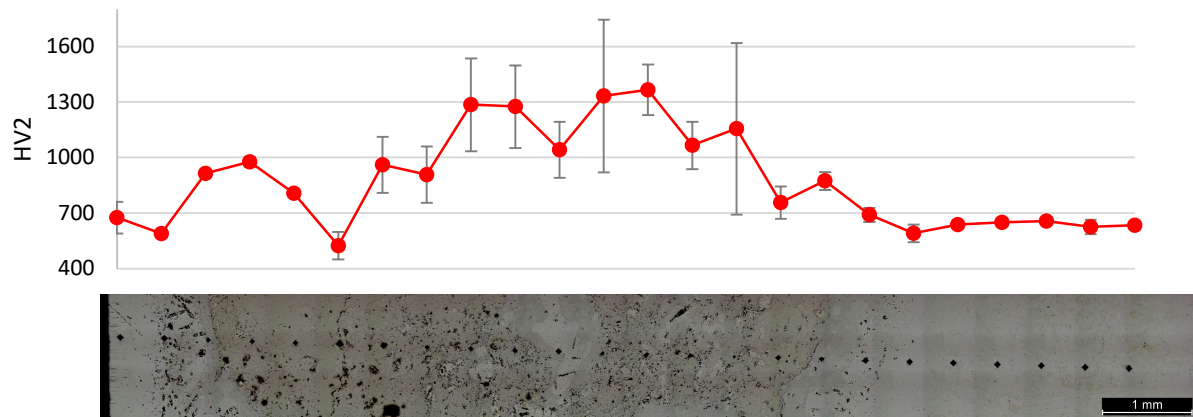


Figure 10 10: Hardness value throughout the profile of the Ti-Ni-B₄C sample.

The interface between the base metal and reinforcement, of the system Ti-Ni-B₄C, **Figure 11:** OM imaging of the interface between reinforcement and base metal in the Ti-Ni-B₄C sample., similarly to what was seen with the Ti-Ni-C, it does not possess a high concentration of porosities or defects in the interface between regions which could cause a discontinuity. The transition between the reinforcement and the base metal has a reaction zone that improves the bonding.

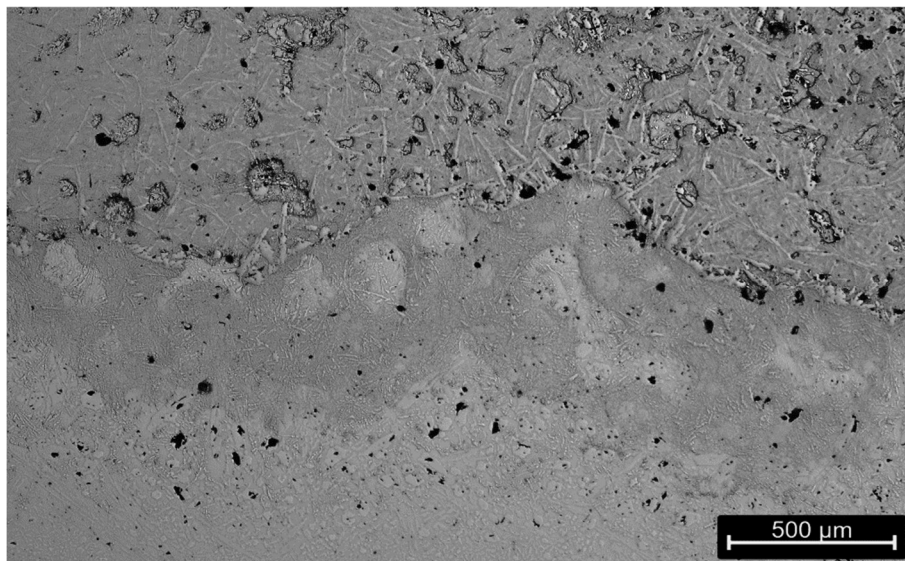


Figure 11: OM imaging of the interface between reinforcement and base metal in the Ti-Ni-B₄C sample.

Analyzing the bulk of reinforcement, **Figure 12**, it possesses a heterogeneous microstructure composed of regions of base metal and regions with TiC particles and TiB₂,

these regions of bare metal are round and larger than in the other samples and are lined TiC and TiB₂ particle, these could originate from the process, Olejnik et al. (Olejnik et al. 2011) first described, of formation of “bubbles” filled with liquid metal, from single particles merging.

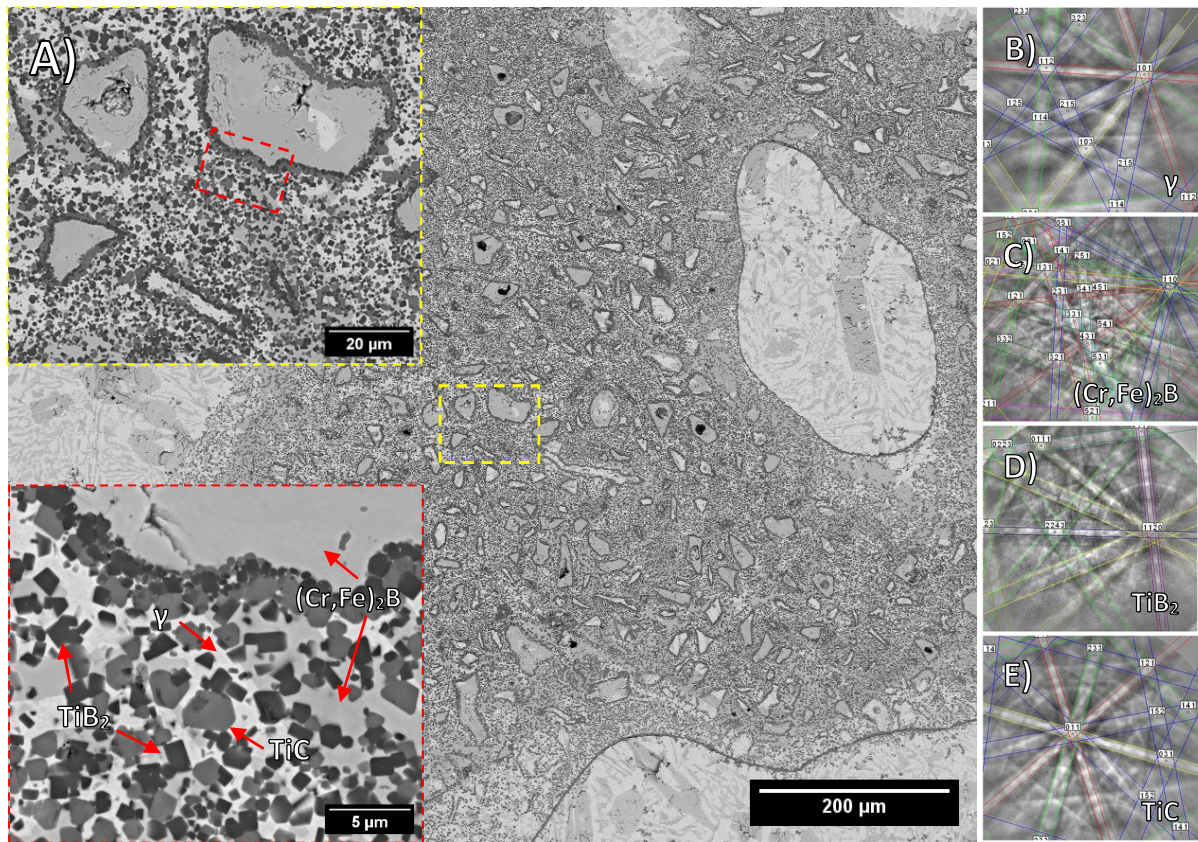


Figure 12: Sample of reinforced HCrWCI with Ti-Ni-B₄C; A) Microstructure of the reinforcement with back-scattered electrons; B) Kikuchi pattern of γ ; C) Kikuchi pattern of $(\text{Cr,Fe})_2\text{B}$; D) Kikuchi pattern of TiB_2 ; E) Kikuchi pattern of TiC .

In the reinforced regions, the microstructure is composed of angular shaped formation lined with a first layer of TiB_2 and a second of TiC , which are mainly filled with $(\text{Cr,Fe})_2\text{B}$, but occasionally with γ , as seen in **Figure 12**. These angular shaped formations, averaging $17.9 \pm 7.7 \mu\text{m}$, might derive from the incomplete combustion of the system, more specifically from the B_4C particles which tend to stay until the final stages of combustion, as illustrated by Yang et al. (Y.F. Yang, Wang, Zhao, et al. 2008), and after the reaction has yielded, dissolved into the liquid metal and $(\text{Cr,Fe})_2\text{B}$. The presence of B_4C in the liquid metal, alters the precipitation of the eutectic $\text{L} \rightarrow \text{M}_7\text{C}_3 + \gamma$, to the peritectic $\text{L} \rightarrow \text{M}_2\text{B} + \gamma$ (Rodríguez-Sebastián 2010). The TiC and TiB_2 particles evenly dispersed in the matrix between the angular shaped formation, average $1.59 \pm 0.58 \mu\text{m}$ and $0.87 \pm 0.38 \mu\text{m}$ respectively.

Looking into the EDS mapping of the elements shown in **Figure 13**, Ti, **Figure 13 B**, can be found both in TiC and TiB_2 particles. It is to be noted that Ti is more prevalent, in the EDS mapping, in the TiC particles when compared to TiB_2 particles, as expected, due to the molar ratios of 1 to 1 and 1 to 2 respectively. Fe, **Figure 13 C**, is present at similar concentrations, dissolved in the matrix, present in $(\text{Cr,Fe})_2\text{B}$. Ni, **Figure 13 D**, is mostly dissolved in the matrix contributing to the γ stabilization. Cr, **Figure 13 E**, can be found mostly in $(\text{Cr,Fe})_2\text{B}$. C, **Figure 13 F**, due to the difficulty of mapping it seems to be only prevalent in the TiC particles, but it is also expected to be present in $(\text{Cr,Fe})_2\text{B}$ and also dispersed in the matrix. Si, **Figure 13 G**, as seen in other systems is dispersed throughout all phases and at a higher concentration in the matrix.

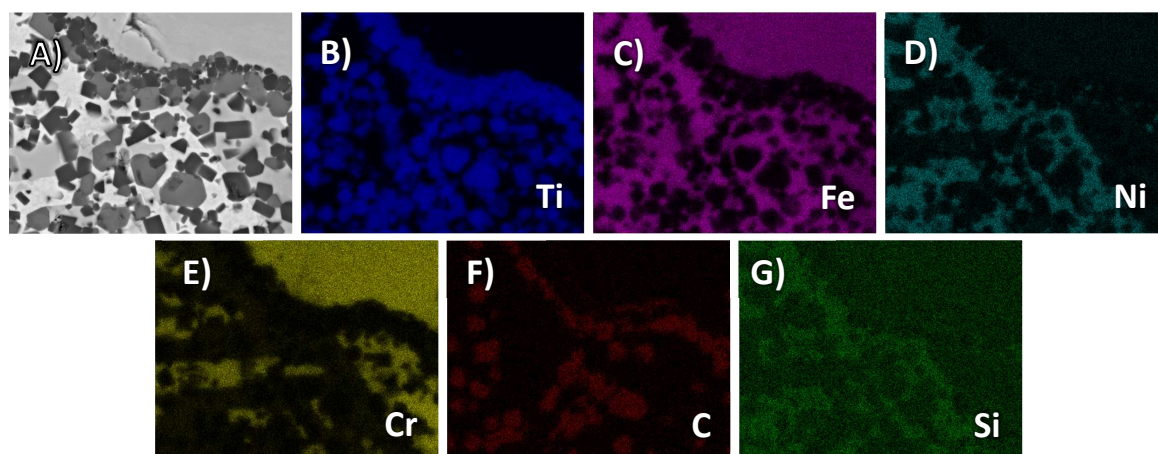


Figure 13 13: A) Back-scattered electron image; B) Ti EDS mapping; C) Fe EDS mapping; D) Ni EDS mapping; E) Cr EDS mapping; F) C EDS mapping; G) Si EDS mapping.

4. Conclusions

The study demonstrates that reinforcing high-chromium white cast iron (HCrWCI) with different systems significantly improves hardness. The base metal exhibited a hardness of 624 ± 26 HV30, whereas the reinforced regions displayed higher hardness values: Ti-Al-C (892 ± 59 HV30), Ti-Ni-C (973 ± 55 HV30), and Ti-Ni-B₄C (965 ± 65 HV30).

The bonding between the reinforcement and the base metal varied among the systems. The Ti-Al-C system exhibited a brittle alumina layer with significant porosity at the interface, potentially compromising its structural integrity. In contrast, the Ti-Ni-C and Ti-Ni-B₄C systems showed a reaction zone with fewer porosities and better bonding, suggesting improved mechanical performance and durability.

The microstructural analysis revealed that the reinforcements altered the matrix and carbide phases. The Ti-Al-C system formed a martensitic matrix with TiC particles and M₇C₃ carbides, while the Ti-Ni-C and Ti-Ni-B₄C systems resulted in an austenitic matrix due to the presence of nickel, which stabilized the γ -phase. In the Ti-Ni-C system M₇C₃ and TiC particles were seen while in the Ti-Ni-B₄C besides TiC particles, TiB₂ particles were also present and the secondary phase (Cr,Fe)₂B. The distribution of reinforcement particles was heterogeneous. In the Ti-Ni-C system, TiC particles tended to cluster, forming a "sponge-like" structure, whereas in the Ti-Al-C system, TiC particles were more uniformly dispersed.

Further research should focus on optimizing the compacting pressure and thermal treatments to minimize unreinforced layers and improve the homogeneity of the reinforcement distribution. Additionally, investigating other reinforcement systems and their long-term performance under various wear and corrosion conditions could provide broader insights into the material's potential applications.

This work highlights the significant advancements in enhancing the mechanical properties of HCrWCI through targeted reinforcements, paving the way for more durable and wear-resistant materials in industrial applications.

References

- ASTM. 2019. *Standard Specification for Abrasion-Resistant Cast Irons*. ASTM.
- Bedolla-Jacuinde, A., F. V. Guerra, A. J. Guerrero-Pastran, M. A. Sierra-Cetina, et S. Valdez-Medina. 2021. « Microstructural effect and wear performance of high chromium white cast iron modified with high boron contents. » *Wear* 476. <https://doi.org/10.1016/j.wear.2021.203675>.
- Carpenter, S. D., D. Carpenter, et J. T. H. Pearce. 2004. « XRD and electron microscope study of an as-cast 26.6% chromium white iron microstructure. » *Materials Chemistry and Physics* 85 (1): 32-40. <https://doi.org/10.1016/j.matchemphys.2003.11.037>.
- Choi, Yoon, et Shi Woo Rhee. 1993. « Effect of aluminium addition on the combustion reaction of titanium and carbon to form TiC. » *Journal of Materials Science* 28 (24): 6669-75. <https://doi.org/10.1007/BF00356413>. <https://doi.org/10.1007/BF00356413>.
- Chung, R. J., X. Tang, D. Y. Li, B. Hinckley, et K. Dolman. 2013. « Microstructure refinement of hypereutectic high Cr cast irons using hard carbide-forming elements for improved wear resistance. » *Wear* 301 (1-2): 695-706. <https://doi.org/10.1016/j.wear.2013.01.079>.
- Doğan, Ö N., J. A. Hawk, et G. Laird. 1997. « Solidification structure and abrasion resistance of high chromium white irons. » *Metallurgical and Materials Transactions A* 28 (6): 1315-28. <https://doi.org/10.1007/s11661-997-0267-3>. <https://dx.doi.org/10.1007/s11661-997-0267-3>.
- Fashu, Simbarashe, et Vera Trabadelo. 2023. « Development and Performance of High Chromium White Cast Irons (HCWCIs) for Wear–Corrosive Environments: A Critical Review. » *Metals* 13 (11). <https://doi.org/10.3390/met13111831>.
- Gonzalez-Pociño, Alejandro, Florentino Alvarez-Antolin, et Juan Asensio-Lozano. 2019. « Erosive Wear Resistance Regarding Different Destabilization Heat Treatments of Austenite in High Chromium White Cast Iron, Alloyed with Mo. » *Metals* 9 (5). <https://doi.org/10.3390/met9050522>.
- ISO. 2023. *Metallic materials - Vickers hardness test*. International Organization for Standardization.
- Jie-Cai, Han, Xing-Hong Zhang, et J. V. Wood. 2000. « In-situ combustion synthesis and densification of TiC–xNi cermets. » *Materials Science and Engineering: A* 280 (2): 328-33. [https://doi.org/https://doi.org/10.1016/S0921-5093\(99\)00606-1](https://doi.org/https://doi.org/10.1016/S0921-5093(99)00606-1). <https://www.sciencedirect.com/science/article/pii/S0921509399006061>.
- Karantzalis, E., A. Lekatou, et H. Mavros. 2013. « Microstructure and properties of high chromium cast irons: effect of heat treatments and alloying additions. » *International Journal of Cast Metals Research* 22 (6): 448-56. <https://doi.org/10.1179/174313309x436637>.
- Kopyciński, D., et S. Piasny. 2012. « Influence of Tungsten and Titanium on the Structure of Chromium Cast Iron. » *Archives of Foundry Engineering* (No 1). <https://doi.org/10.2478/v10266-012-0011-3>. <http://journals.pan.pl/Content/86272> <http://journals.pan.pl/dlibra/publication/edition/86272>.
- Moreira, A. B., L. M. M. Ribeiro, P. Lacerda, A. M. P. Pinto, et M. F. Vieira. 2022. « A Study on a Cast Steel Reinforced with WC-Metal Matrix Composite. » *Materials (Basel)* 15 (18). <https://doi.org/10.3390/ma15186199>. <https://www.ncbi.nlm.nih.gov/pubmed/36143506>.

- Moreira, A. B., L. M. M. Ribeiro, P. Lacerda, R. O. Sousa, A. M. P. Pinto, et M. F. Vieira. 2020. « Preparation and Microstructural Characterization of a High-Cr White Cast Iron Reinforced with WC Particles. » *Materials (Basel)* 13 (11). <https://doi.org/10.3390/ma13112596>.
<https://www.ncbi.nlm.nih.gov/pubmed/32517285>.
- Moreira, A. B., R. O. Sousa, P. Lacerda, L. M. M. Ribeiro, A. M. P. Pinto, et M. F. Vieira. 2020. « Microstructural Characterization of TiC-White Cast-Iron Composites Fabricated by In Situ Technique. » *Materials (Basel)* 13 (1). <https://doi.org/10.3390/ma13010209>.
<https://www.ncbi.nlm.nih.gov/pubmed/31947831>.
- Olejniak, Ewa, Andrzej Janas, A. Kolbus, et Gabriela Sikora. 2011. « The composition of reaction substrates for TiC carbides synthesis and its influence on the thickness of iron casting composite layer. » *Archives of Foundry Engineering* 11.
- Purba, Riki Hendra, Kazumichi Shimizu, Kenta Kusumoto, Yila Gaqi, et Takayuki Todaka. 2022. « Effect of boron addition on three-body abrasive wear characteristics of high chromium based multi-component white cast iron. » *Materials Chemistry and Physics* 275. <https://doi.org/10.1016/j.matchemphys.2021.125232>.
- Raghavan, V. 2003. « B-Cr-Fe-Ti (Boron-Chromium-Iron-Titanium). » *Journal of Phase Equilibria* 24 (5): 459-60. <https://doi.org/10.1361/105497103770330163>.
<https://doi.org/10.1361/105497103770330163>.
- Rodriguez-Sebastián, Carlos. 2010. *The Fe-Cr-B phase diagram: Application to liquid phase sintering of Boron containing stainless steels*.
- Song, M. S., B. Huang, M. X. Zhang, et J. G. Li. 2009. « Study of formation behavior of TiC ceramic obtained by self-propagating high-temperature synthesis from Al-Ti-C elemental powders. » *International Journal of Refractory Metals and Hard Materials* 27 (3): 584-89. <https://doi.org/10.1016/j.ijrmhm.2008.09.009>.
- Standardization, International Organization for. 2008. *Fine ceramics (advanced ceramics, advanced technical ceramics) - Determination of the abrasion resistance of coatings by a micro-scale abrasion test*. International Organization for Standardization.
- . 2015. *Metallic materials - Instrumented indentation test for hardness and materials parameters, Part 1: Test methods*. International Organization for Standardization.
- Tabrett, C. P., I. R. Sare, et M. R. Ghomashchi. 1996. « Microstructure-property relationships in high chromium white iron alloys. » *International Materials Reviews* 41 (2): 59-82. <https://doi.org/10.1179/imr.1996.41.2.59>.
<https://journals.sagepub.com/doi/abs/10.1179/imr.1996.41.2.59>.
- Wang, H. T., S. Q. Zhang, J. H. Huang, J. L. Zhu, H. Zhang, et X. K. Zhao. 2013. « Fabrication of TiC/Fe-Ni cermet coatings by reactive thermal spraying of Fe-Ti-Ni-C composite powder. » *Powder Metallurgy* 54 (3): 231-35. <https://doi.org/10.1179/003258909x12450768326983>.
- Wiengmoon, A., T. Chairuangsi, et J. T. H. Pearce. 2005. « An Unusual Structure of an As-cast 30% Cr Alloy White Iron. » *ISIJ International* 45 (11): 1658-65. <https://doi.org/10.2355/isijinternational.45.1658>.
- Yang, Y. F., H. Y. Wang, R. Y. Zhao, Y. H. Liang, et Q. C. Jiang. 2008. « Effect of Ni content on the reaction behaviors of self-propagating high-temperature synthesis in the Ni-Ti-B4C system. » *International Journal of Refractory Metals and Hard Materials* 26 (2): 77-83. <https://doi.org/10.1016/j.ijrmhm.2007.03.001>.

- Yang, Ya-feng, Hui-yuan Wang, Yun-hong Liang, Ru-yi Zhao, et Qi-chuan Jiang. 2007. « Fabrication of steel matrix composites locally reinforced with different ratios of TiC/TiB₂ particulates using SHS reactions of Ni-Ti-B₄C and Ni-Ti-B₄C-C systems during casting. » *Materials Science and Engineering: A* 445-446: 398-404. <https://doi.org/10.1016/j.msea.2006.09.062>.
- . 2008. « Effect of C particle size on the porous formation of TiC particulate locally reinforced steel matrix composites via the SHS reaction of Ni-Ti-C system during casting. » *Materials Science and Engineering: A* 474 (1-2): 355-62. <https://doi.org/10.1016/j.msea.2007.04.061>.
- Zhu, G., W. Wang, R. Wang, C. Zhao, W. Pan, H. Huang, D. Du, D. Wang, D. Shu, A. Dong, B. Sun, S. Jiang, et Y. Pu. 2017. « Formation Mechanism of Spherical TiC in Ni-Ti-C System during Combustion Synthesis. » *Materials (Basel)* 10 (9). <https://doi.org/10.3390/ma10091007>. <https://www.ncbi.nlm.nih.gov/pubmed/28850088>.
- Zou, Binglin, Ping Shen, Xueqiang Cao, et Qichuan Jiang. 2012. « The mechanism of thermal explosion (TE) synthesis of TiC-TiB₂ particulate locally reinforced steel matrix composites from an Al-Ti-B₄C system via a TE-casting route. » *Materials Chemistry and Physics* 132 (1): 51-62. <https://doi.org/10.1016/j.matchemphys.2011.10.051>.

Acknowledgments

The authors would like to thank the contributions given by Dinis Carneiro and Mariana Alves in research and metallographic preparation.